ME101 INTRO SESSION

Yeditepe University Society of Mechanical Engineering

YUSOME Club Overview & The Future of Mechanical Engineering

Organization, Concrete 2025–2026 Activities, ME101 Course Learning Outcomes, Top 10 Future Engineering Trends & Interactive Kahoot

PRESENTED BY	INTERACTIVE EVALUATION	TOTAL DURATION TARGET
YUSOME	10-Question Kahoot	Max 50 Minutes Total

OUR TEAM

YUSOME Organizational Structure & Active Recruitment



Altay Mağıltaş

President

Strategic vision & ops



Kerem Ayyıldız

Vice President

Strategic vision & ops



Efe Nalbant

Secretary General

Institutional administration



Eren Ertürk

Treasurer (Sayman)

Budget & financial planning



Ece İrem Tamer

Recording Secretary (Yazman)

Documentation & official records

Apply Today

Join YUSOME as an Active Committee Member!

Gain real project management experience, organize major summits, and build your network.

OUR TEAM

YUSOME Organizational Structure & Active Recruitment



Ayşe Alkan

Supervisory Board Chair

Governance oversight



Ali Gökalp Tıltak

Supervisory Board Member

Governance oversight



Mustafa Melih Erdem

Supervisory Board Member

Governance oversight



Beyza Yıldız

Promotion & Design Committee

Visual identity, social media management, brand outreach.



Ece Oral

Event & Organization Committee

Summit coordination, technical trip logistics, workshops.

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INDUSTRIAL EXPOSURE

2025–2026 Technical Field Trips

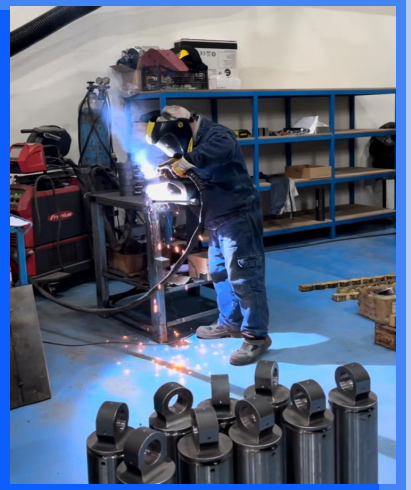
Single-day local and intercity plant visits connecting engineering students directly with active production lines and practicing engineers.



Vestas
Renewable Energy & Wind Turbines

ASELSAN
Defense Electronics & Advanced Tech





Atlaslift & Fenese
Gas Springs & Hydraulic Systems
Armored Vehicles & Defense Systems

HKTM
Motion Control & Industrial Automation



ME101 LO Alignment: LO 5 (Global Challenges), LO 9 (UN Sustainability), LO 2 (Ethics)

FLAGSHIP SUMMIT

MechaniCareer Conference 2025–2026









MechaniCareer'2026

A full-day flagship summit featuring engineering executives, technical keynotes, panel sessions, and 1-on-1 career Q&A opportunities.

2025–2026 Participating Industry Keynotes:

TEI Aero-engines	**Ford Otosan** Automotive R&D	**Mercedes** Vehicle Engineering
Bayer Process Tech	**Şişecam** Materials & Glass	**ARUP** Global Engineering

🎓 **ME101 LO Alignment: LO 10 (Technical Presentation), LO 12 (Innovation & Entrepreneurship)**

SOCIAL & COMMUNITY

Social Workshops & Social Responsibility Seminars

SOCIAL BONDING

New Year Cookie & Clay Workshops

High-attendance social workshops designed to build cross-year friendships and reduce midterm stress.



SOCIAL RESPONSIBILITY

International Day of People with Disabilities Seminar

Hosted 3 educators from Ataşehir RAM on special needs education and neurodiversity in engineering.



ACADEMIC ALIGNMENT

Direct Alignment with ME101 Learning Outcomes (LOs)

LO 13: Teamwork

Executing summits & trips through committee teamwork and peer collaboration.

LO 2: Engineering Ethics

Understanding social responsibility, workplace safety, and neurodiversity awareness.

LO 9: UN SDGs

Clean energy exposure via Vestas wind power and green manufacturing trips.

LO 5: Global Challenges

Evaluating global engineering shifts with TEI, Ford Otosan, and Mercedes leaders.

LO 12: Entrepreneurship

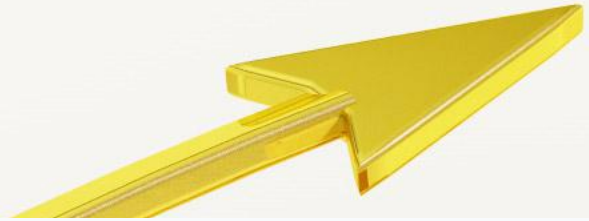
Fostering innovation mindset and direct networking at MechaniCareer.

LO 4: CAD & Tech Skills

Peer-led technical workshops and 3D modeling support sessions.

FUTURE OUTLOOK

Top 10 Future Trends in Mechanical Engineering (1–5)

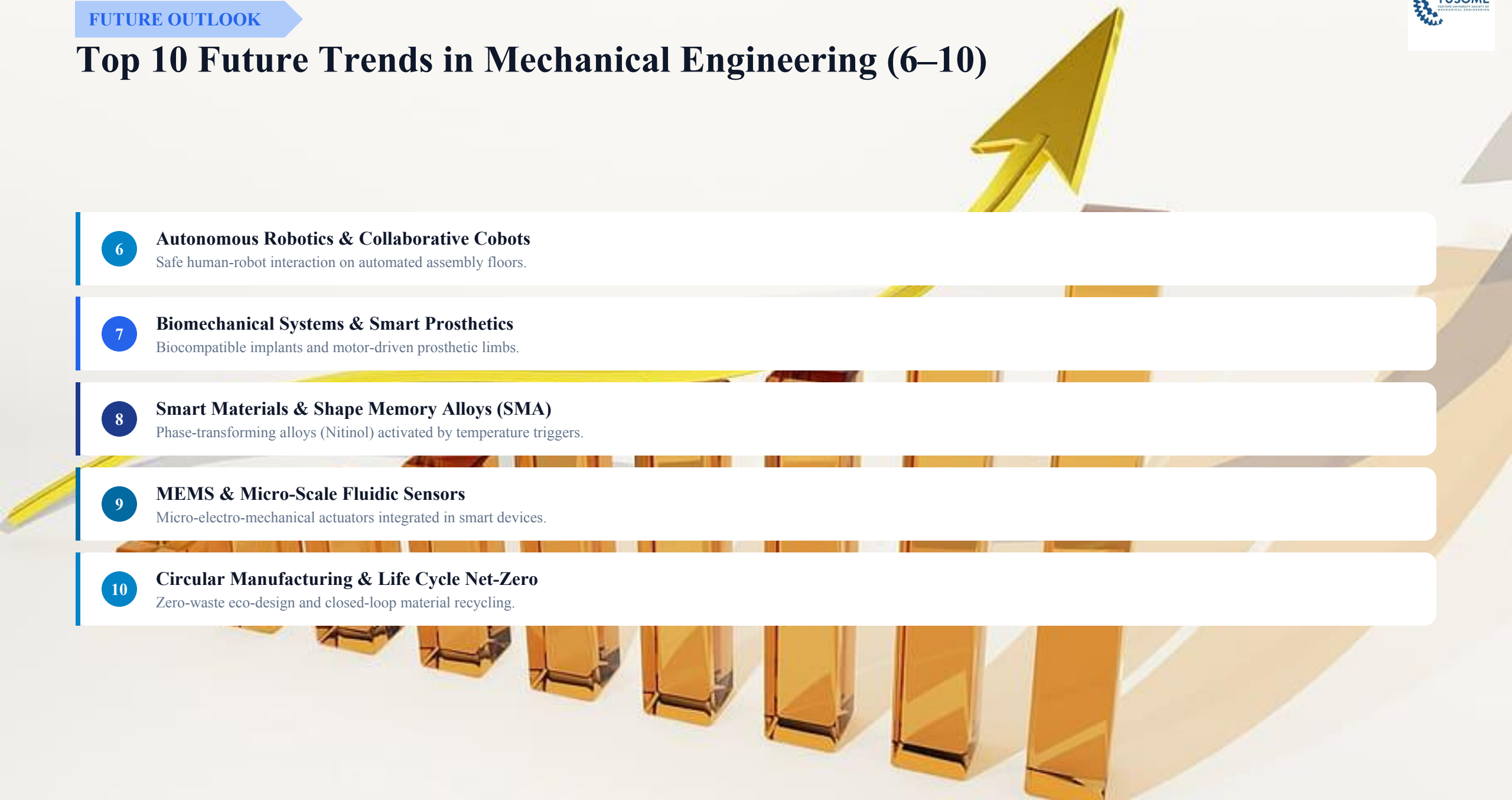


1. **AI-Driven Generative Design & Topology Optimization**
   Algorithmic material reduction for ultra-lightweight aerospace structures.
2. **Green Hydrogen & Clean Energy Propulsion**
   Zero-emission fuel cell systems and renewable power storage.
3. **Metal Additive Manufacturing (Industrial 3D/4D Printing)**
   Laser powder bed fusion for complex biomedical & turbine components.
4. **Digital Twin & Real-Time Predictive Health Monitoring**
   Virtual simulation models coupled with live IoT sensor telemetry.
5. **EV Battery Thermal Management & Micro-Cooling**
   Micro-channel liquid cooling to prevent thermal runaway in high-density batteries.



FUTURE OUTLOOK

Top 10 Future Trends in Mechanical Engineering (6–10)



6 **Autonomous Robotics & Collaborative Cobots**
Safe human-robot interaction on automated assembly floors.

7 **Biomechanical Systems & Smart Prosthetics**
Biocompatible implants and motor-driven prosthetic limbs.

8 **Smart Materials & Shape Memory Alloys (SMA)**
Phase-transforming alloys (Nitinol) activated by temperature triggers.

9 **MEMS & Micro-Scale Fluidic Sensors**
Micro-electro-mechanical actuators integrated in smart devices.

10 **Circular Manufacturing & Life Cycle Net-Zero**
Zero-waste eco-design and closed-loop material recycling.

AI-Driven Generative Design & Topology Optimization

How can a part become lighter without becoming too weak?

Set the problem

Engineers define loads, materials and design limits.

Explore better shapes

Software generates options; topology optimization removes unnecessary material.

The next challenge

Make lighter parts that can also be manufactured and tested.



An optimized aerospace bracket: material follows the loads.

Image: nTop / NASA EXCITE bracket

REMEMBER

Software proposes. Engineers check strength and manufacturability.

Green Hydrogen & Clean Energy Propulsion

Can renewable electricity be stored as a fuel?

Produce and store

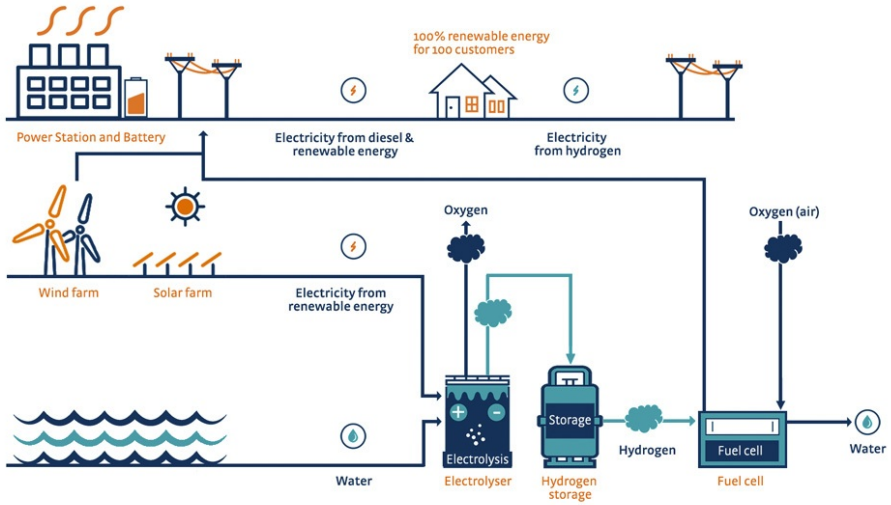
Renewable electricity splits water into hydrogen and oxygen.

Use the hydrogen

A fuel cell produces electricity, water and heat.

The next challenge

Design efficient tanks, compressors and cooling; reduce cost and losses.



A renewable hydrogen demonstration system in Denham, Australia.

Image: ARENA / Horizon Power

REMEMBER

Hydrogen is an energy carrier; its production route matters.

Metal Additive Manufacturing Industrial 3D Printing

How do we make a curved cooling channel inside one metal part?

Build layer by layer

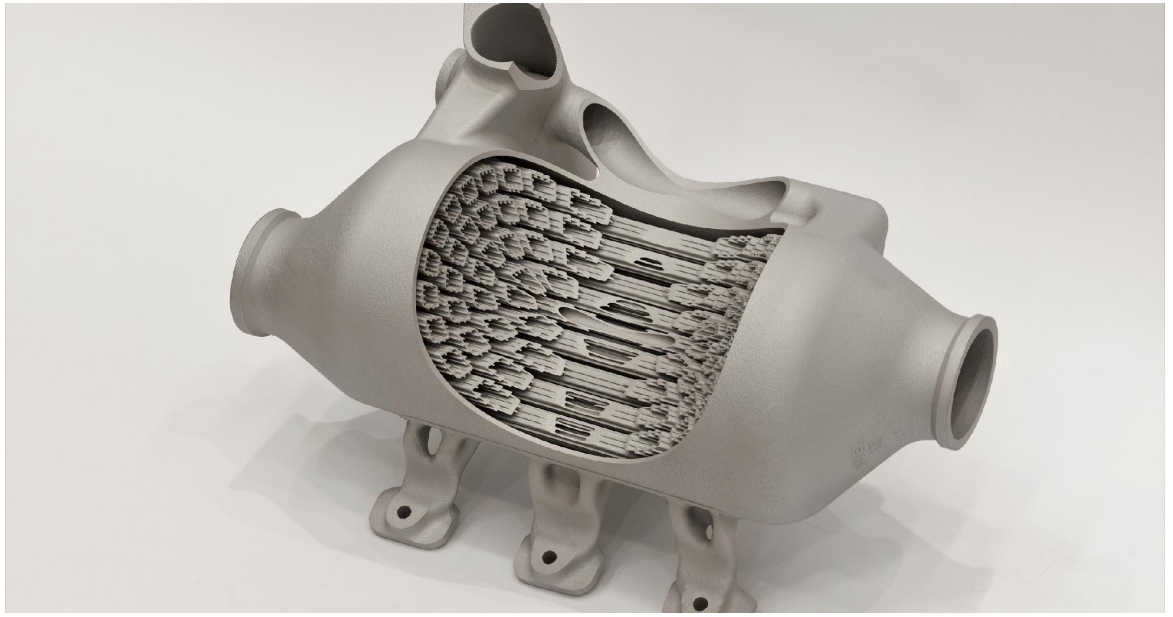
For example, a laser melts selected areas of metal powder.

Create internal geometry

Complex cooling channels can be made in a single build.

The next challenge

Improve repeatable quality; inspect defects and finish surfaces.



A heat-exchanger design with complex internal flow passages.

Image: NVIDIA / Sintavia

REMEMBER

A key advantage: complex internal channels inside one part.

Digital Twins & Predictive Maintenance

Can we notice a problem before a machine fails?

Connect the real and virtual

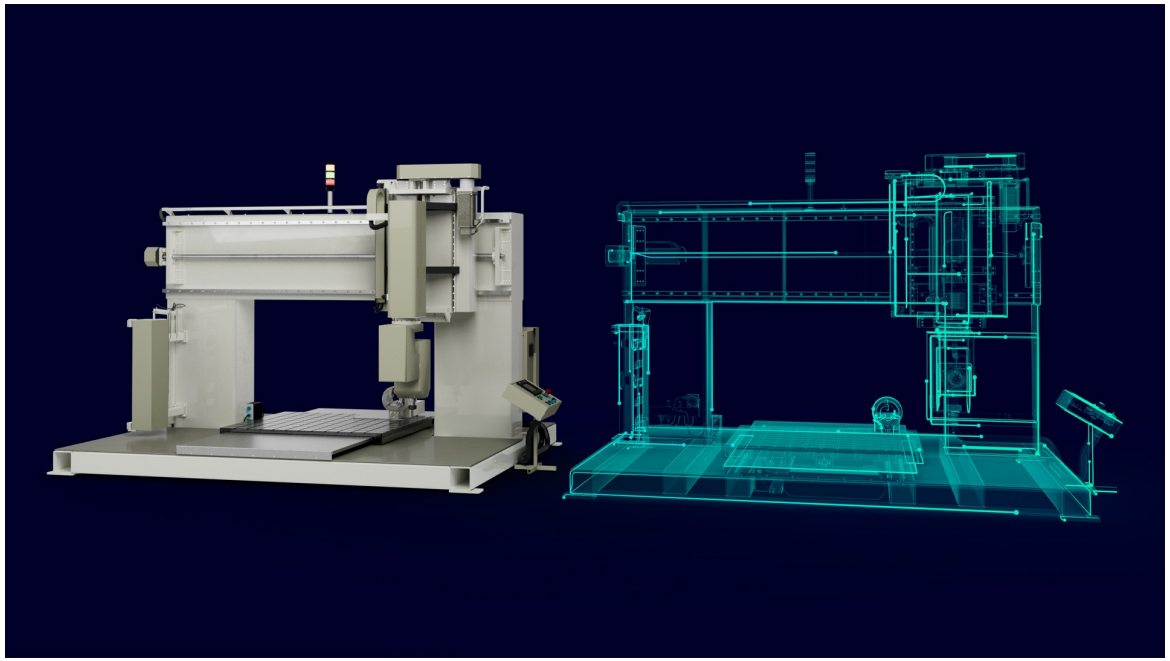
Live sensors update a digital model of a physical system.

Use the measurements

Temperature and vibration data help track machine condition.

The next challenge

Use reliable data and validated models to plan maintenance.



A real machine and its virtual representation; sensor data links them.

Image: Siemens

REMEMBER

Physical system + live sensor data + an updated virtual model.

EV Battery Thermal Management & Advanced Cooling

Why does an electric car still need a cooling system?

Keep cells in the right range

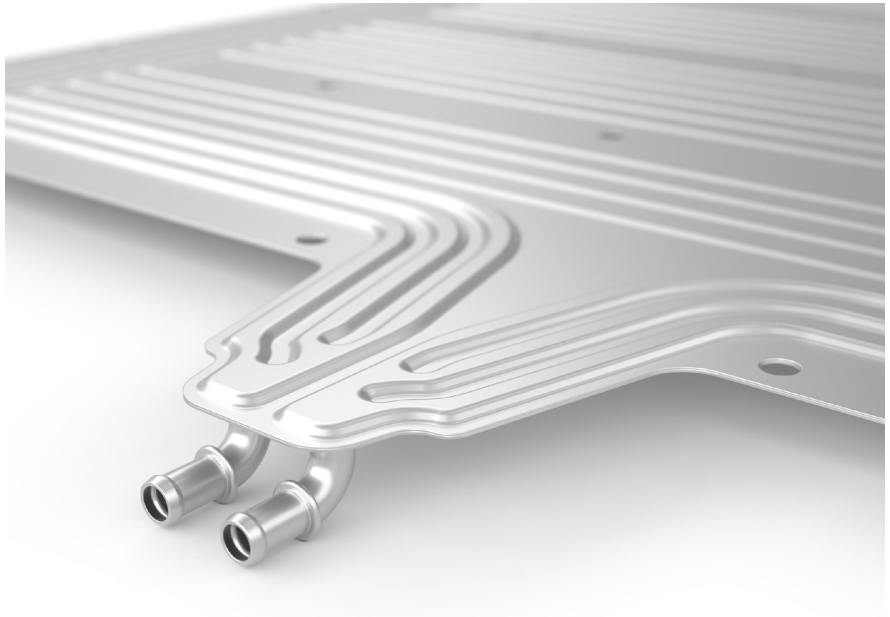
Batteries generate heat during driving and charging.

Control hot and cold spots

Cooling plates remove heat; cold weather may require heating.

The next challenge

Support fast charging while limiting weight, energy use and risk.



A battery cooling plate carries heat away through coolant channels.

Image: MAHLE

REMEMBER

Keep cells at a suitable, even temperature for safety and performance.

Autonomous Robots & Collaborative Cobots

How can a robot share a workspace with a person?

Two different ideas

Autonomy means acting with less direct control; collaboration means working with people.

Manage contact and distance

Safety functions can limit force and speed or trigger a protective stop.

The next challenge

Make the whole task safe, including tools, parts and people.



A cobot assists a worker in a shared production environment.

Image: Universal Robots / Nichrominox

REMEMBER

A cobot needs suitable safety functions and a task risk assessment.

Biomechanical Systems & Smart Prosthetics

How can an artificial limb respond to its user?

Sense the user and movement

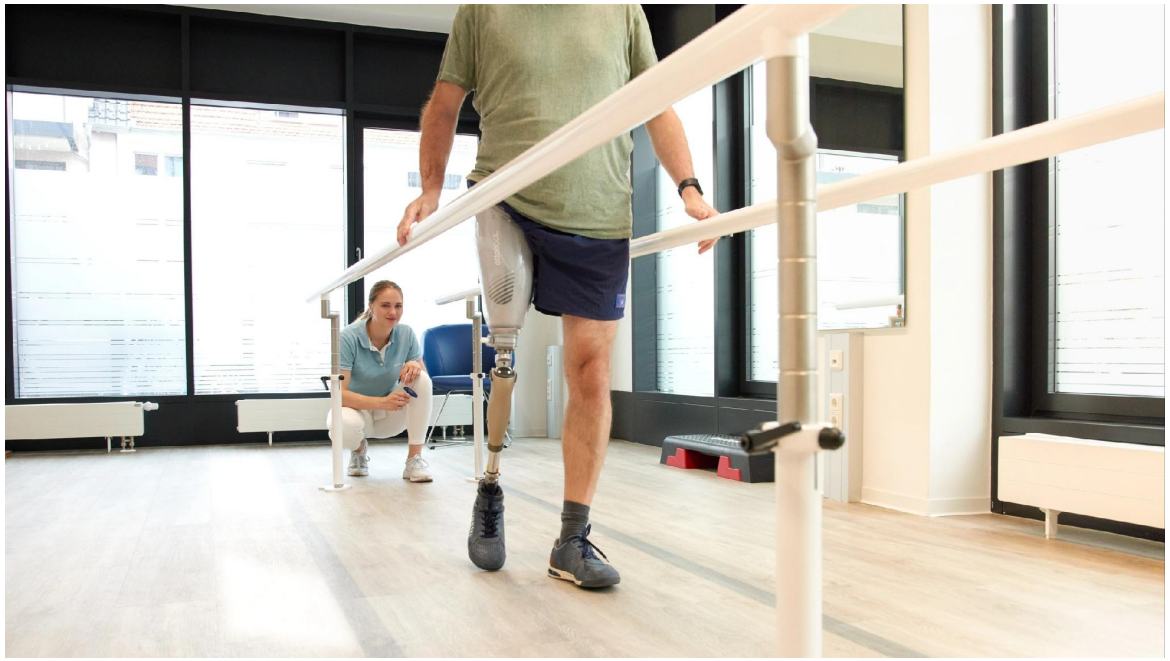
Sensors may measure muscle signals, force or joint motion.

Control a mechanical response

A controller adjusts motors or resistance in the prosthetic joint.

The next challenge

Improve comfort, natural movement, durability and affordability.



Prosthetic gait training: fit and real-world movement matter.

Image: Ottobock

REMEMBER

Smart prosthetics combine mechanics, sensing and control.

Smart Materials & Shape Memory Alloys

Can a metal wire remember a shape?

A material example: Nitinol

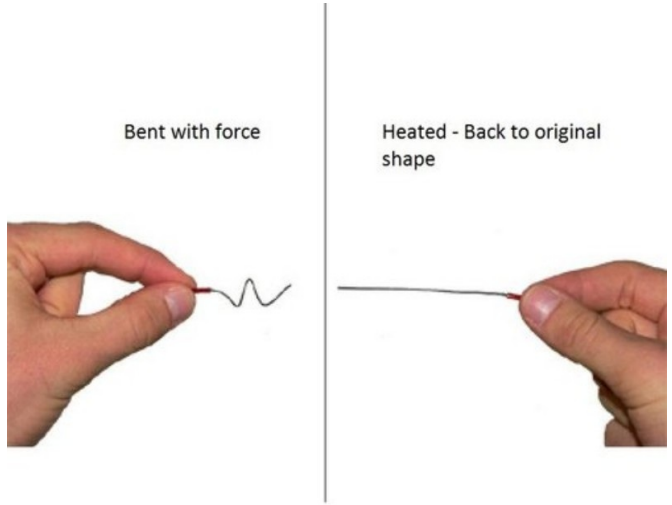
This nickel–titanium alloy can have a preset shape.

Heat triggers shape recovery

After suitable deformation, heating can restore that preset shape.

The next challenge

Design repeatable actuators; manage fatigue and heating–cooling time.



Nitinol wire: bent at low temperature, then heated to recover its shape.

Image: XOT Metals

REMEMBER

Heat restores the preset shape; electric current can supply that heat.

MEMS & Microfluidic Sensors

Where is the mechanical engineering inside a tiny sensor?

MEMS: tiny mechanics

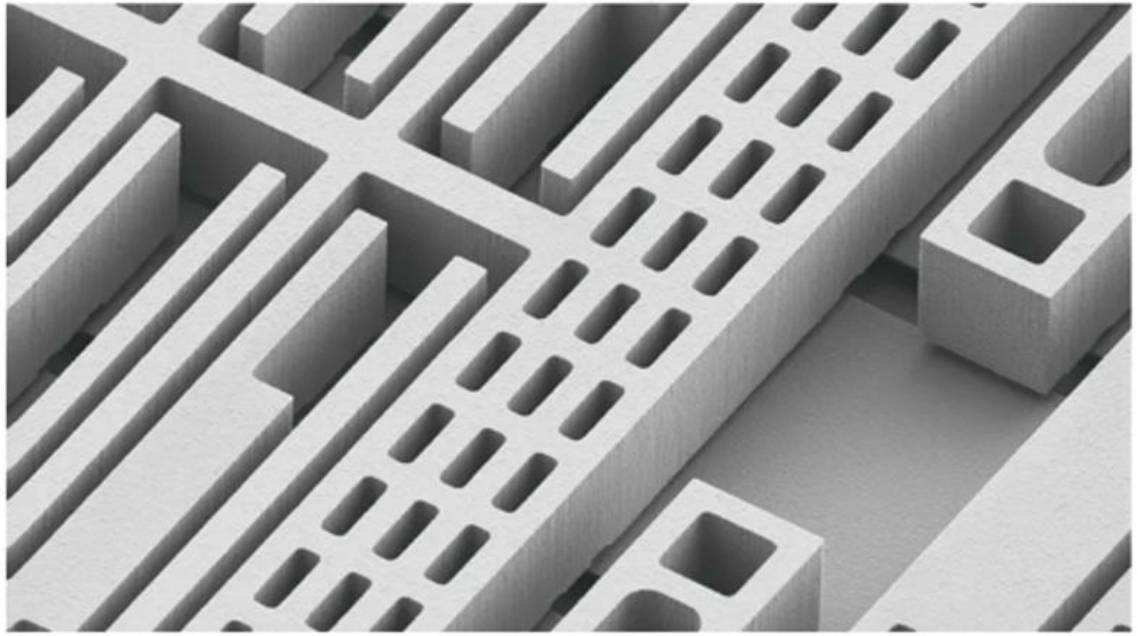
Micro-electro-mechanical systems combine small moving structures and electronics.

Microfluidics: tiny flows

Small channels handle very small liquid samples for measurement.

The next challenge

Improve accuracy, packaging and resistance to contamination.



Inside a MEMS sensor: microscopic mechanical structures on a chip.

Image: Bosch Sensortec

REMEMBER

Small size, familiar physics: motion, pressure and fluid flow.

Circular Manufacturing & Life-Cycle Emissions

What if a worn product became the starting point for the next one?

Keep products useful

Design for repair, reuse, disassembly and remanufacturing.

Measure the whole life cycle

Include materials, production, use and end-of-life treatment.

The next challenge

Recover value while measuring real resource and emission savings.



Remanufacturing restores a used component for another service life.

Image: Caterpillar / Cat Reman

REMEMBER

Repair and reuse first where practical; circular does not automatically mean net zero.

Kahoot Quiz Breakdown - 5 Points Total

3

Club Questions

Testing attentiveness on YUSOME activities & team structure.

7

Future & Technical ME

Differentiating engineering questions covering future topics.

0.5

Points per Question

Tanışma Toplantısı

- 21 Ekim, Çarşamba
- 17.30
- The Co
- ***Hepinizi Bekliyoruz!***



GET READY FOR KAHOOT

Thank You! Join YUSOME & Good Luck!

Open Kahoot.it on your mobile devices. Enter the Game PIN displayed on screen!

@yusomeyeditepe LinkedIn: yusome7tepe Committee Applications Open